



National Radio
Astronomy
Observatory

NAASC
NORTH AMERICAN ALMA SCIENCE CENTER

ALMA Proposal Preparation Workshop

March 25, 2026 @INCT-UDA, Copiapó

1.1. Basics of Radio Interferometry

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➤ Radio regime

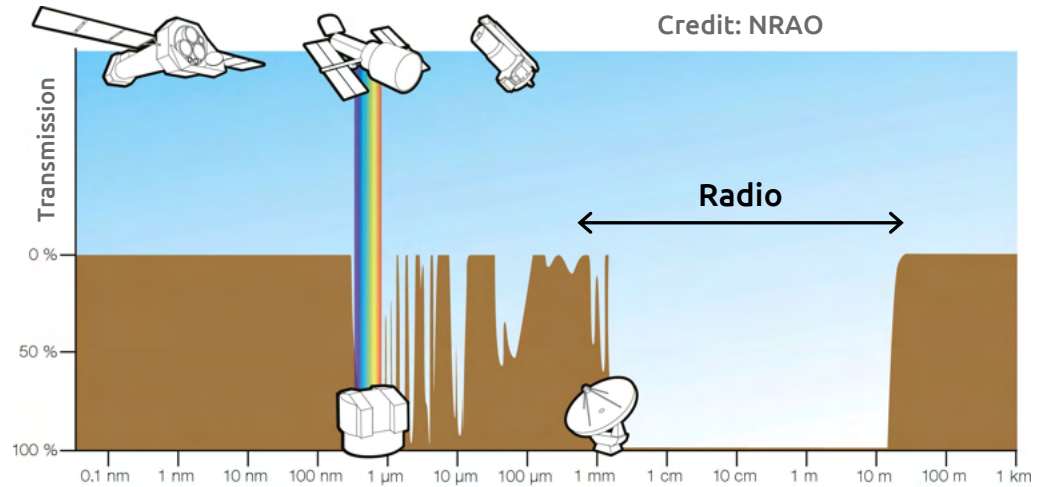
- Submm to meter wavelengths
- Cold, dusty environments
- Complementary to optical/IR
- Jansky: $1 \text{ Jy} = 10^{-26} \text{ W m}^{-2} \text{ Hz}^{-1}$

➤ Physical processes

- Thermal emission from dust and ionized gas
- Spectral lines from atomic and molecular transitions
- Non-thermal processes (synchrotron, free-free cont.)

➤ Angular reso: $\theta_{\text{res}} \propto \lambda/D$

- Long λ implies poor θ_{res} (1 dish)
- Large apertures are required



IRAM (30m)



GBT (100m)

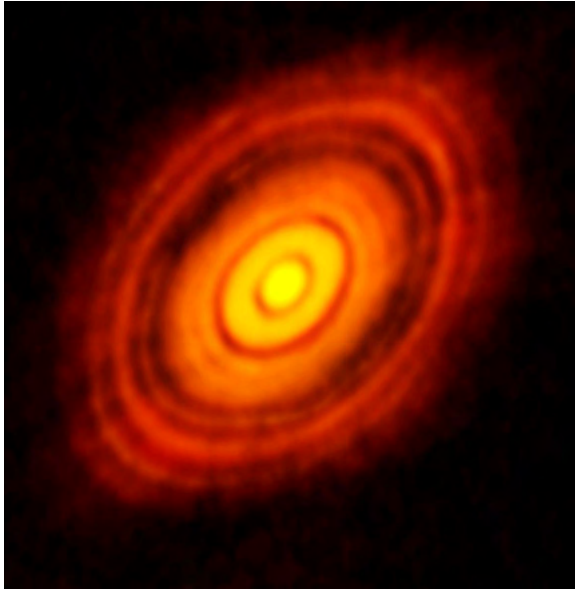


FAST (500m)

Origins of planets

HL Tau protoplanetary disk

ALMA Partnership et al. (2015)

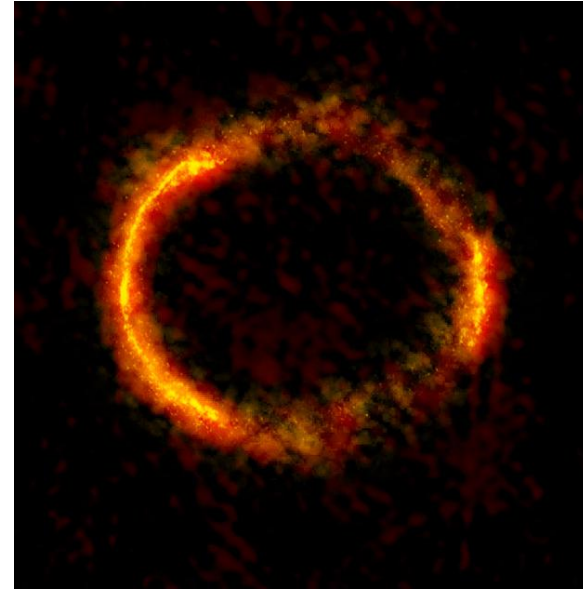


1.4 arcsec

Galaxy evolution

Lensed galaxy SDP.81 at $z = 3$

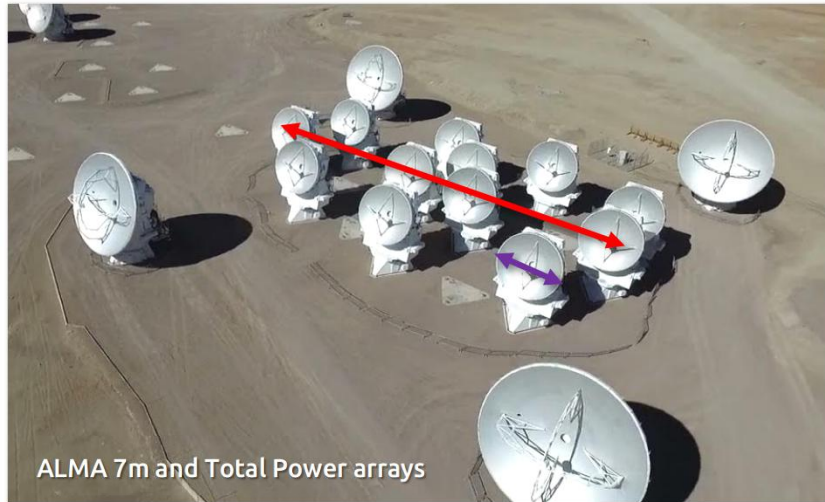
ALMA Partnership et al. (2015)



4 arcsec

- Some science cases require angular resolutions well beyond what single-dish telescopes can provide → use of radio interferometry

- The FOV becomes the diffraction limit of a **single antenna**.
- Resolution of a telescope equivalent in size to the **largest baseline**.



Angular Resolution

$$\theta_{\text{res}} \sim \lambda / B_{\text{max}}$$

Largest distance between antennas

FOV or Primary Beam

$$\theta_{\text{PB}} \sim \lambda / D$$

Diffraction limit of a single antenna

Maximum Recoverable Scale

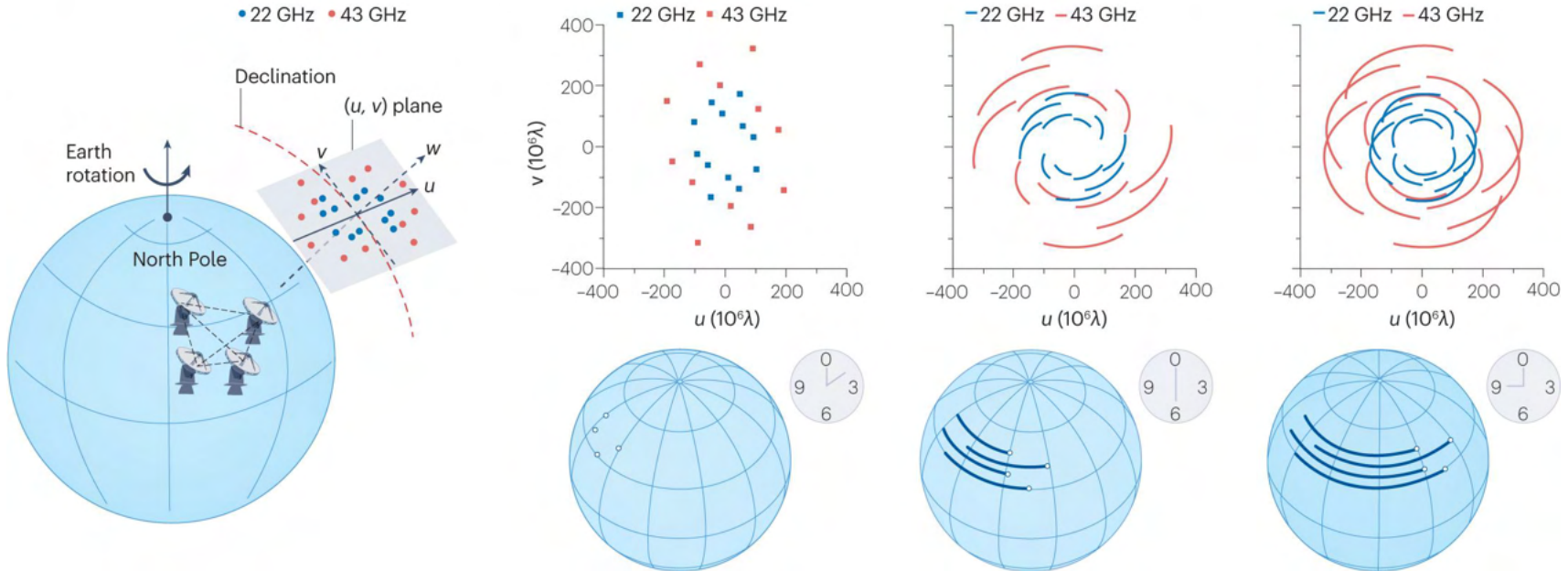
$$\theta_{\text{MRS}} \sim \lambda / B_{\text{min}}$$

Shortest distance between antennas

Radio Interferometry

- Aperture synthesis: separate antennas working as a single telescope
- Data taken in the uv-plane: **Amplitude** (how much) & **Phase** (where)

Aperture Synthesis
Physics Nobel Prize 1974
M. Ryle & A. Hewish



Asaki et al. (2023), *Nature Reviews*

Each pair of antennas samples a point in the uv-plane, and the sky is scanned through Earth rotation and antenna motion.

- Complex visibility in the Fourier domain:

$$V(u, v) = \iint T(l, m) e^{-2\pi i (ul + vm)} dl dm$$

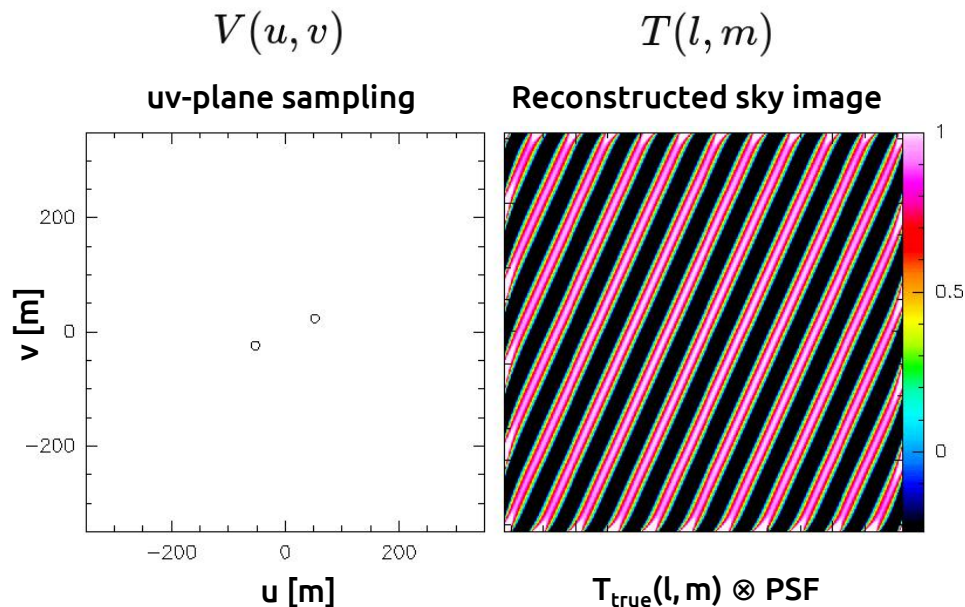
- Sky brightness in the Image domain:

$$T(l, m) = \iint V(u, v) e^{+2\pi i (ul + vm)} du dv$$

- Van Cittert-Zernike theorem:

$$V(u, v) = \text{FT}\{ T(l, m) \}$$

$$T(l, m) = \text{FT}^{-1}\{ V(u, v) \}$$



Credit: NRAO

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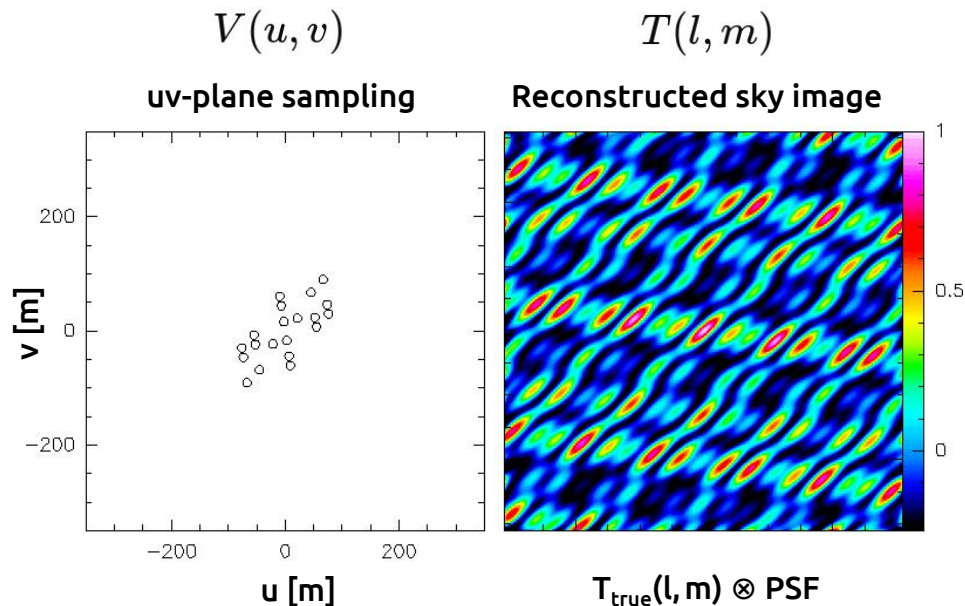
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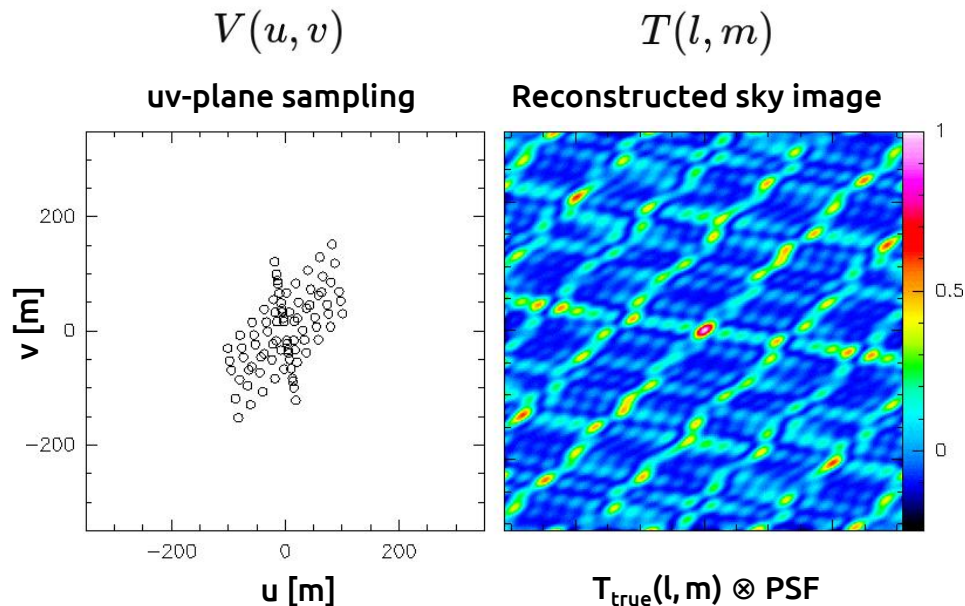
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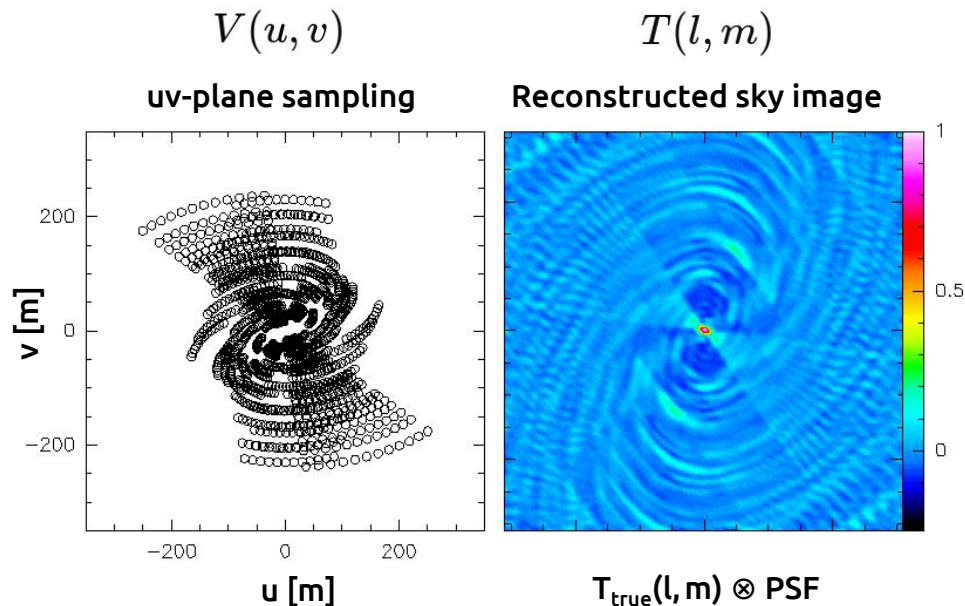
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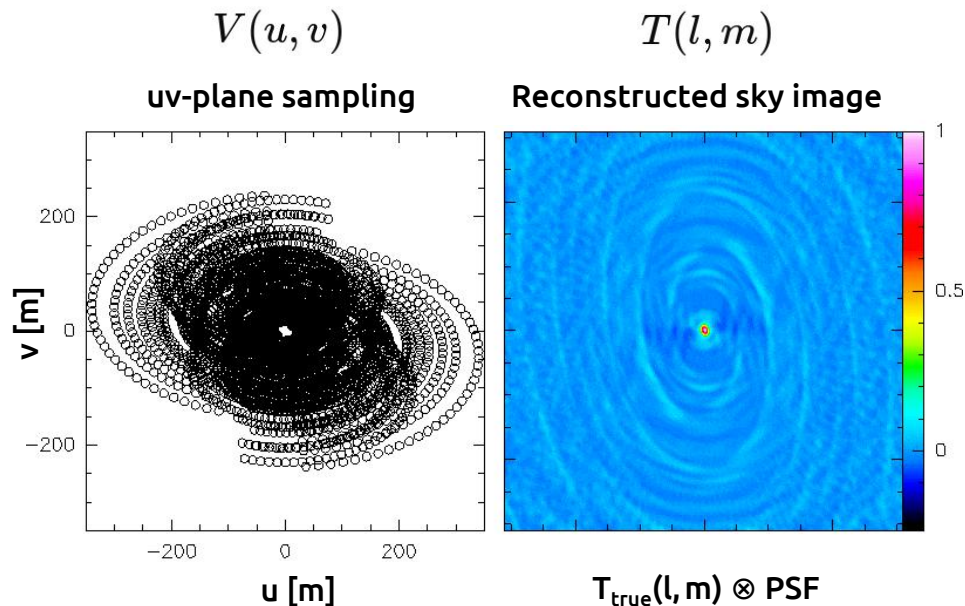
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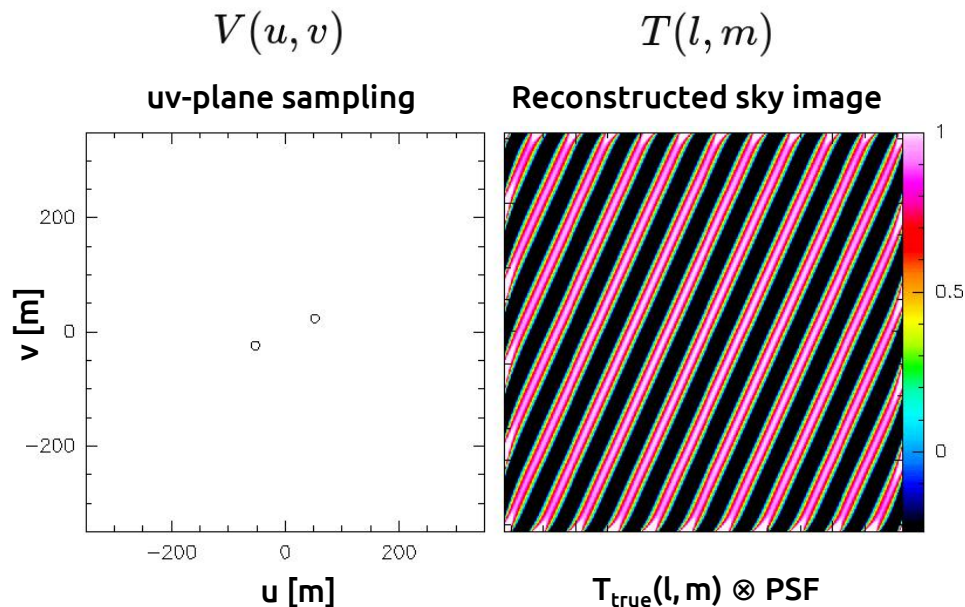
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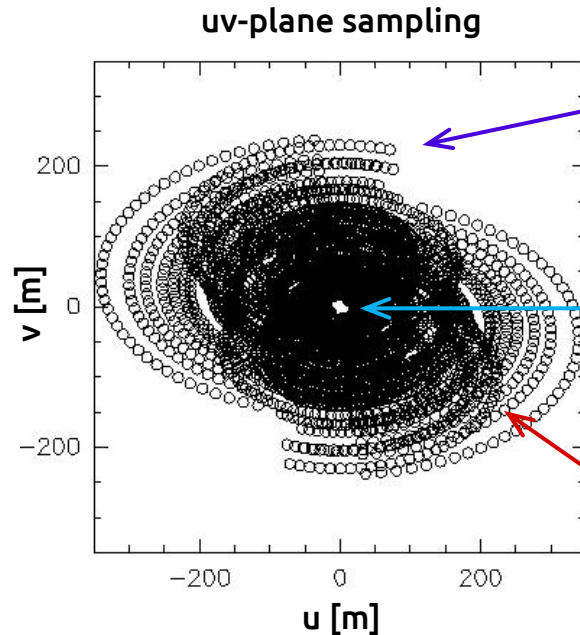
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Credit: NRAO



Outer boundary

Sets the angular resolution limit: $\theta_{\text{res}} = \lambda/B_{\text{max}}$

Inner hole

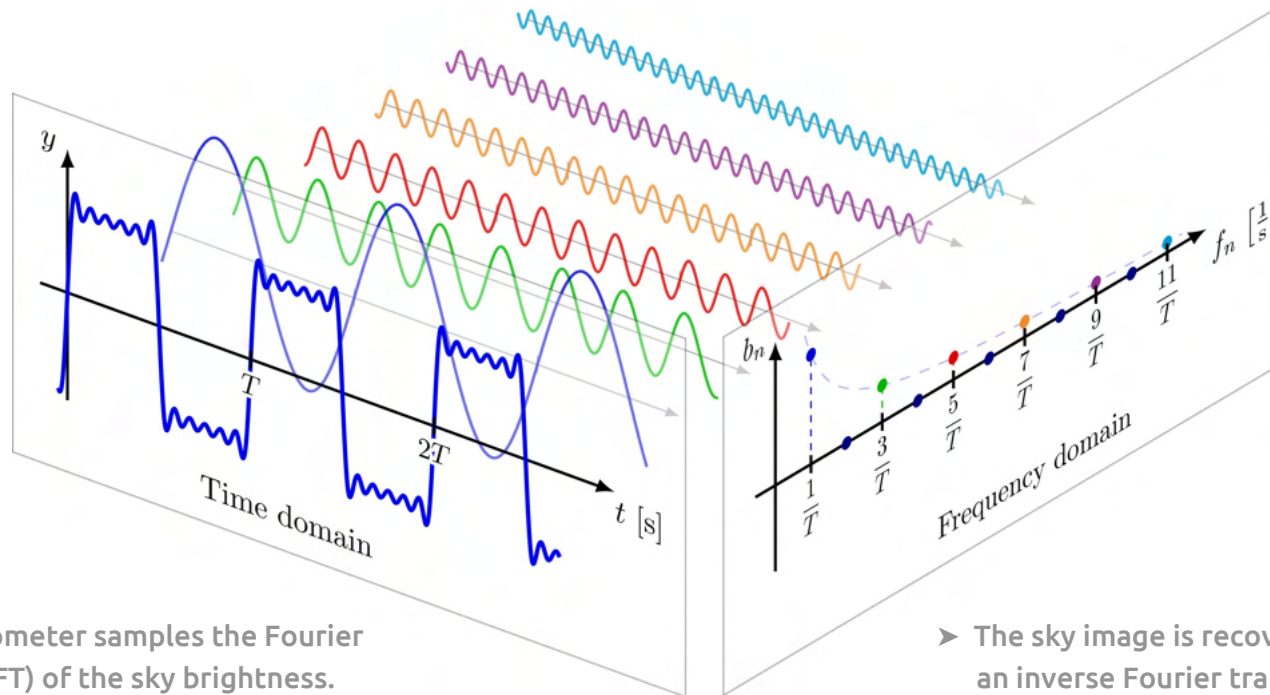
- Insensitive to large angular scales ($\theta > \lambda/B_{\text{min}}$)
- Sets the maximum "recoverable scale"

Incomplete sampling

- Missing info: imperfect image reconstruction
- Creates sidelobes in the synthesized beam

Relation between Visibilities and Sky Brightness

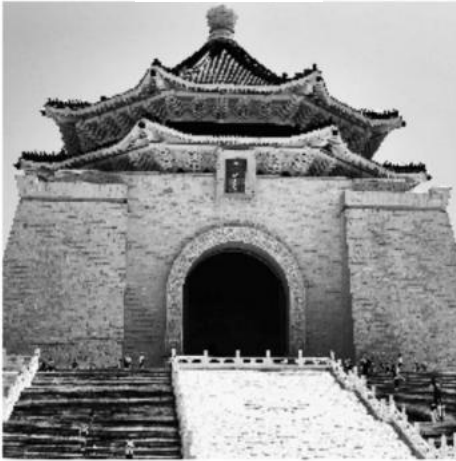
- Sky brightness distribution can be represented as a sum of pure frequencies
- Interferometers measure these Fourier components ($\equiv$ visibilities)



- An interferometer samples the Fourier transform (FT) of the sky brightness.

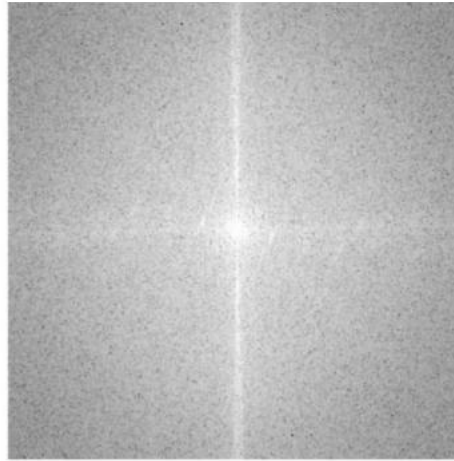
- The sky image is recovered via an inverse Fourier transform.

$I(l, m)$

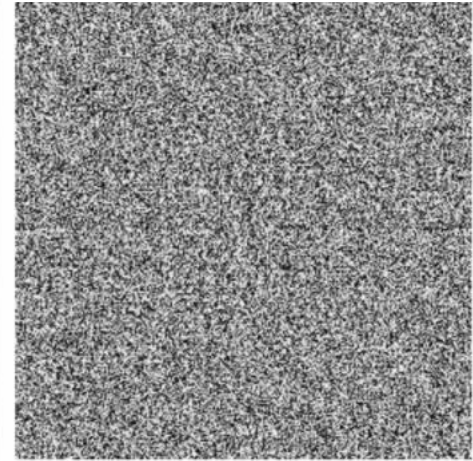


$\mathcal{F}$
 $\Rightarrow$

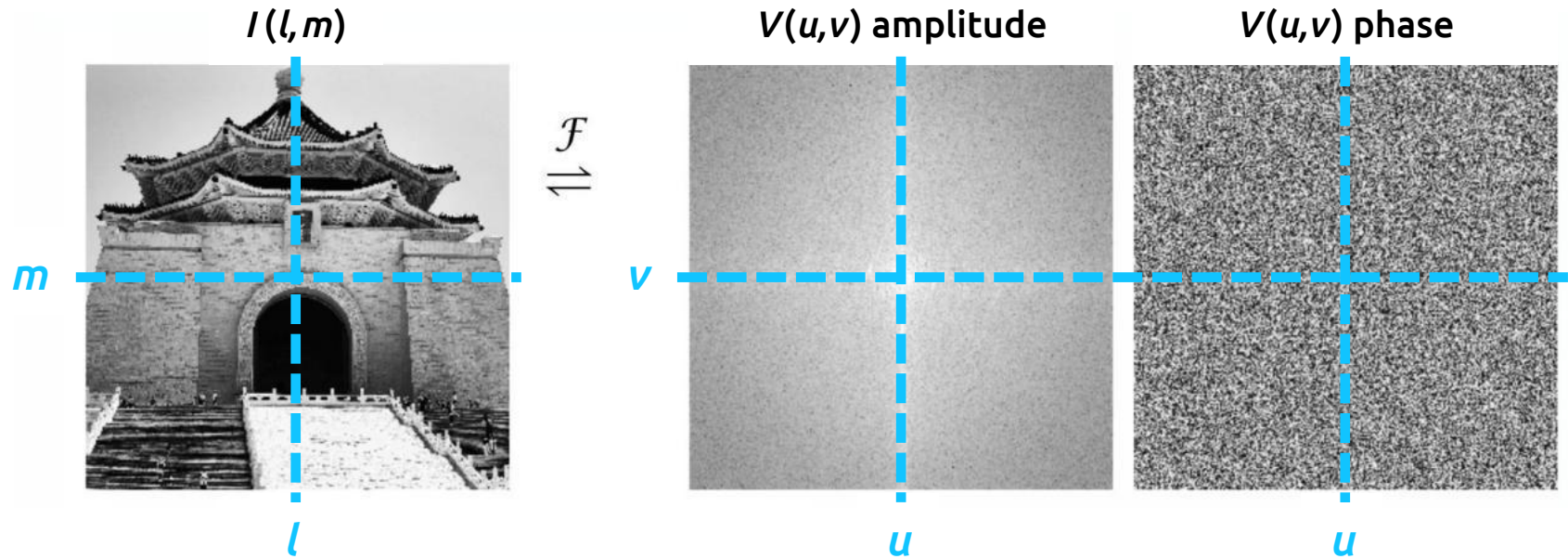
$V(u, v)$ amplitude



$V(u, v)$ phase



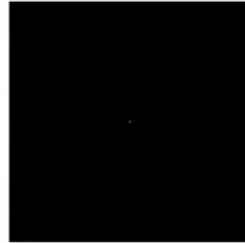
Relation between Visibilities and Sky Brightness



Examples of Fourier Transform Pairs

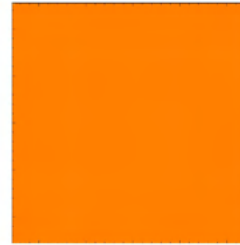
Sky brightness $I(l, m)$

δ Function



$\text{Amp}\{ V(u, v) \}$

Constant



Gaussian



Gaussian



Gaussian

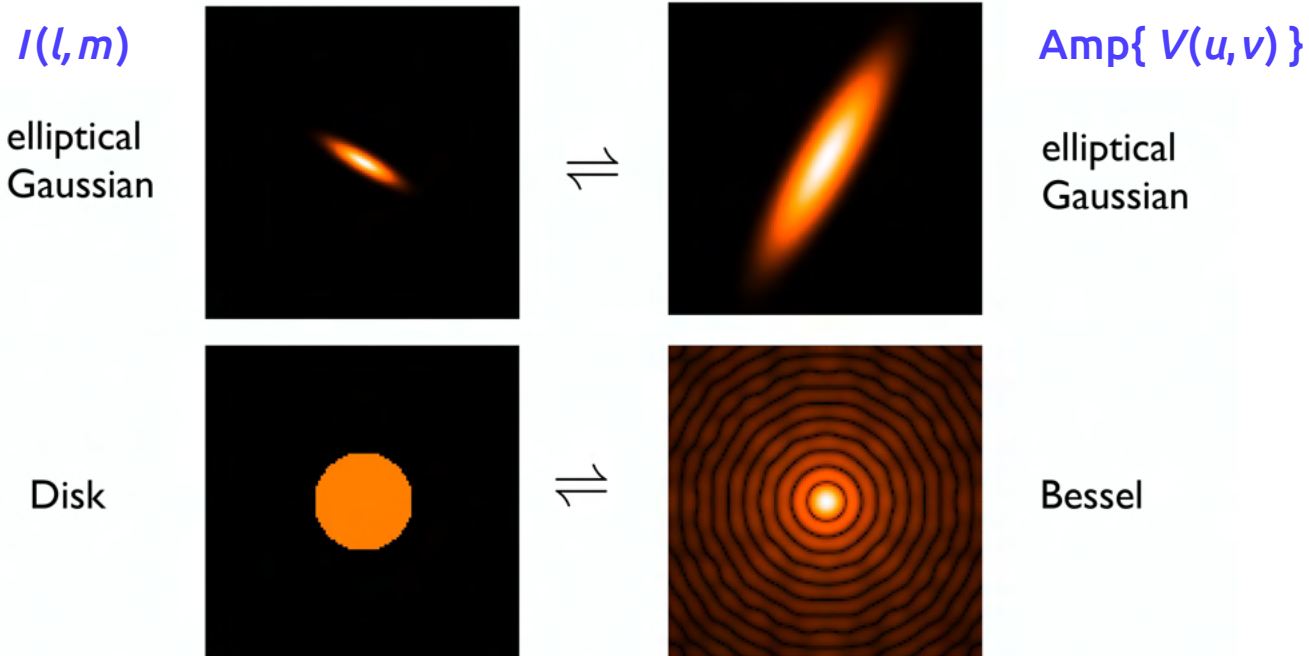


Gaussian



► Narrow features transform to wide features (and vice-versa).

Examples of Fourier Transform Pairs

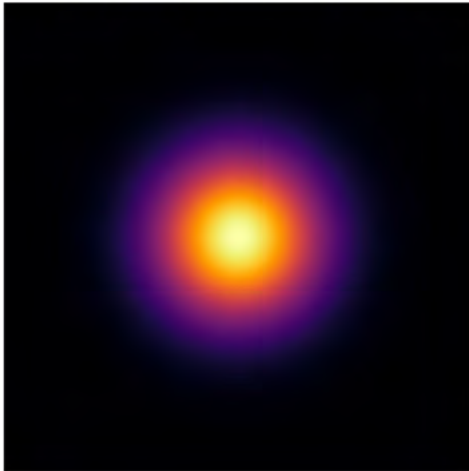


- Sharp edges produce many high spatial frequencies, leading to sinc-like ringing patterns (Gibbs phenomenon).

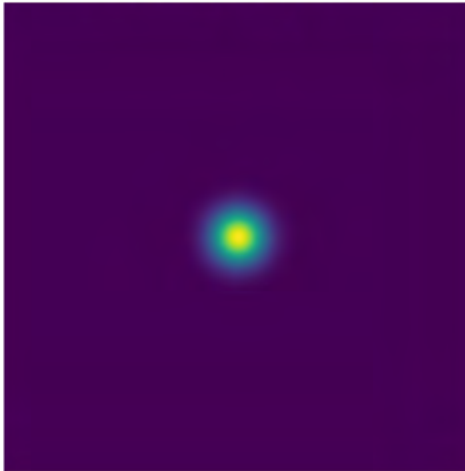
Amplitude and Phase

- Amplitude: **how much** of a frequency component is present.
- Phase: **where** this frequency component is located.

Sky brightness: $I(l,m)$



Visibility amplitude

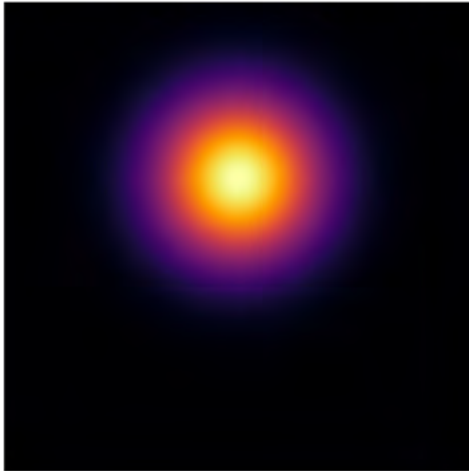


Visibility phase



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Visibility amplitude



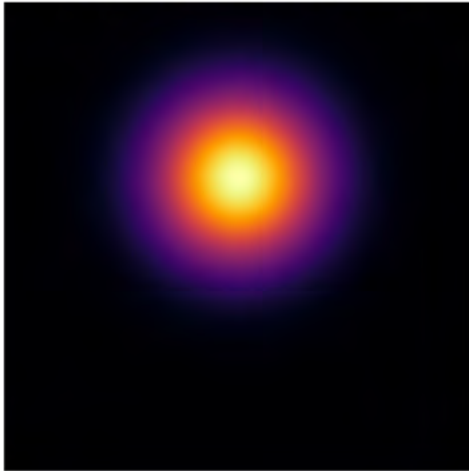
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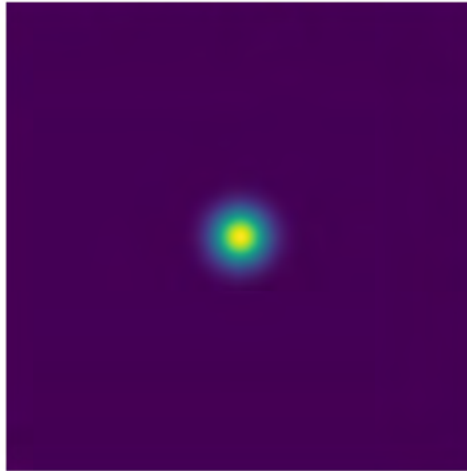
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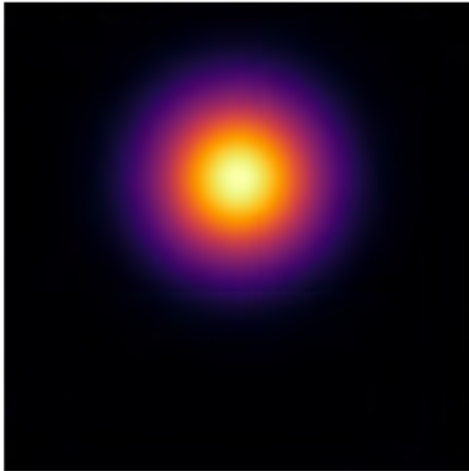


Visibility phase

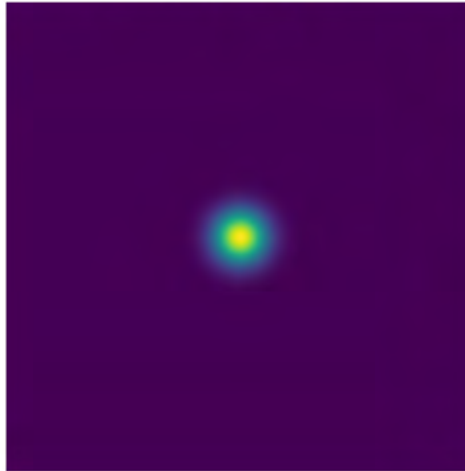


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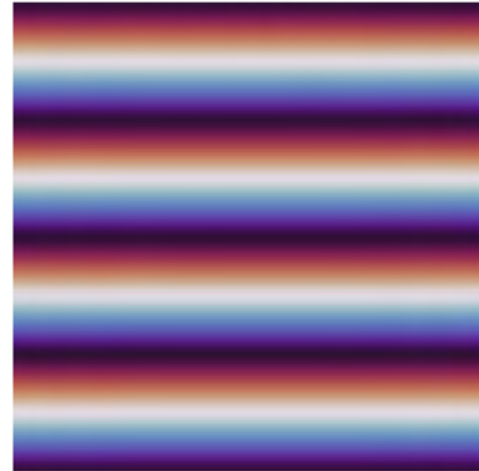
Sky brightness: $I(l, m)$



Visibility amplitude



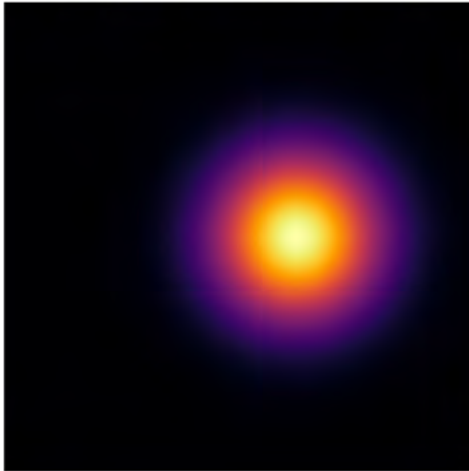
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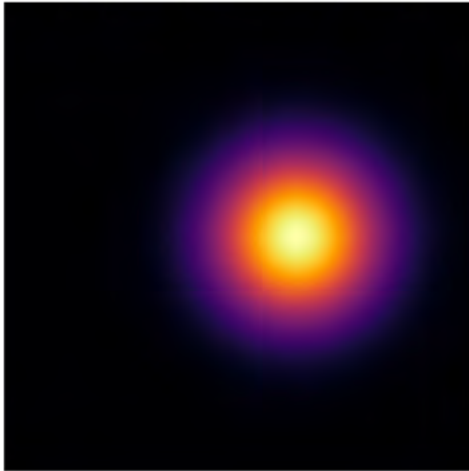
Visibility phase



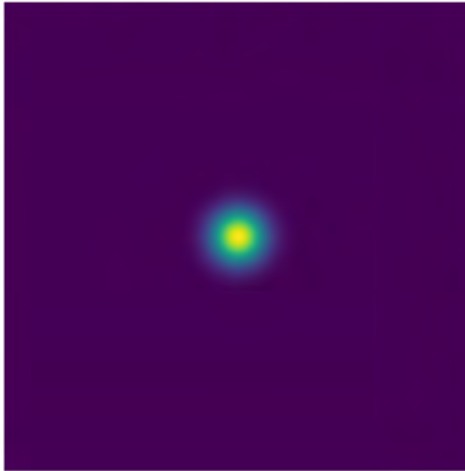
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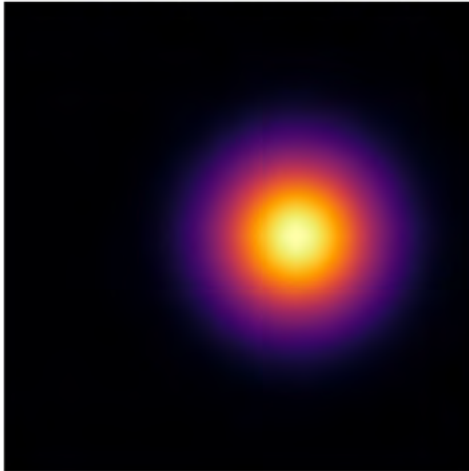
Visibility phase



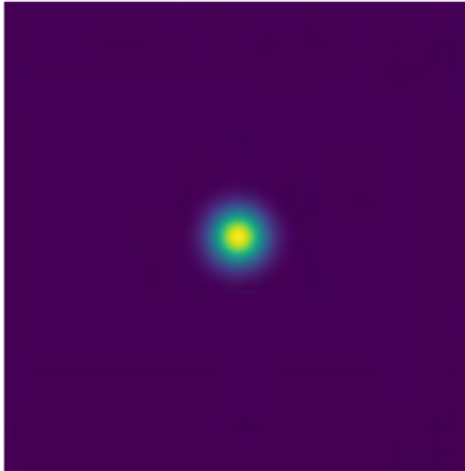
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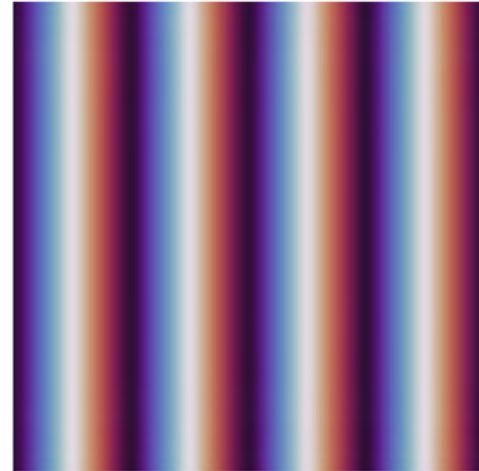
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Visibility amplitude



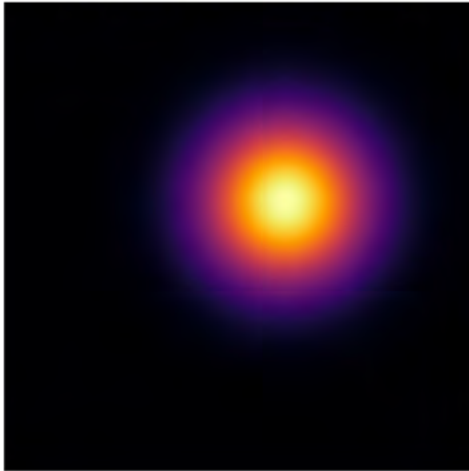
Visibility phase



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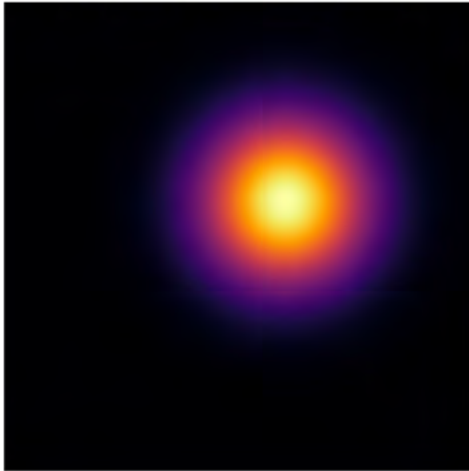
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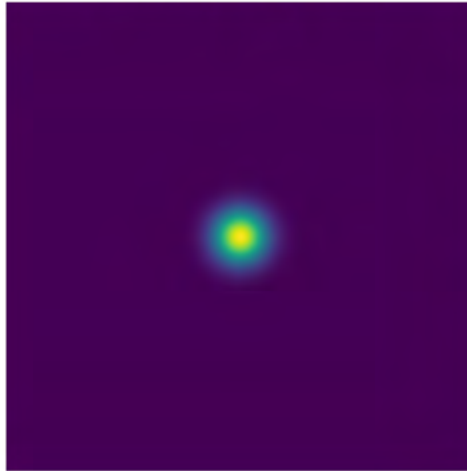
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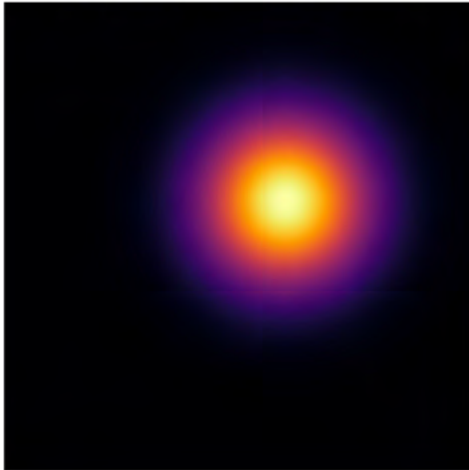


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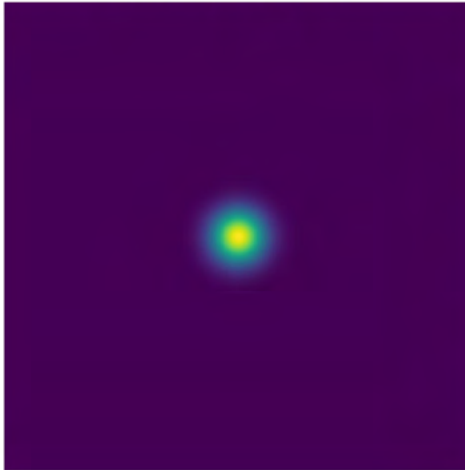


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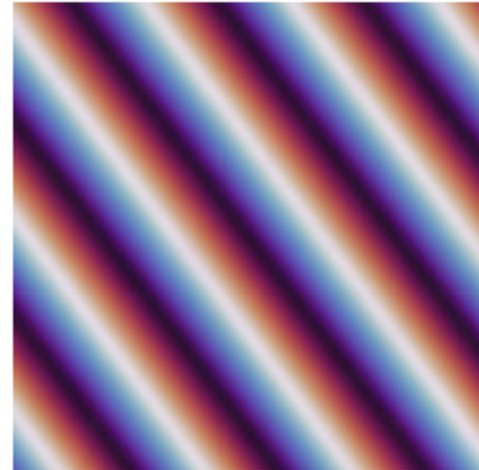
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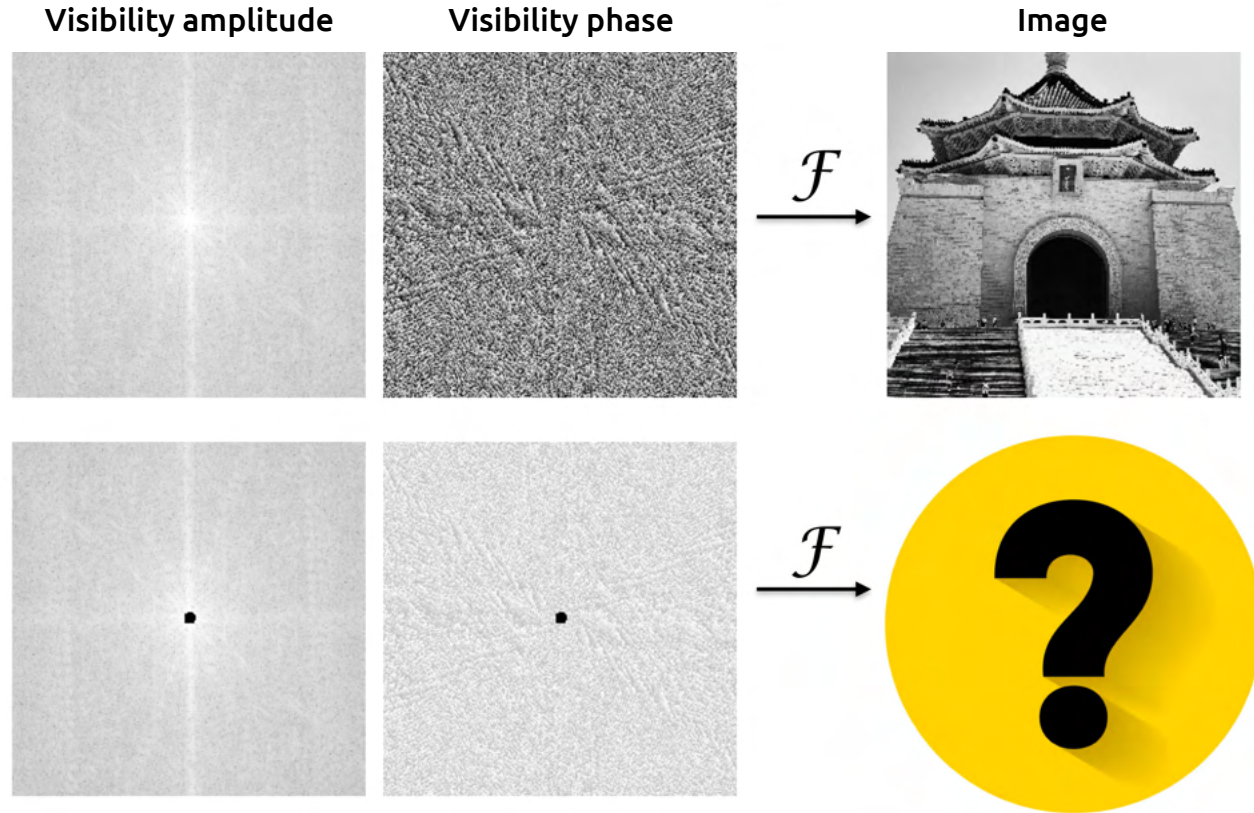
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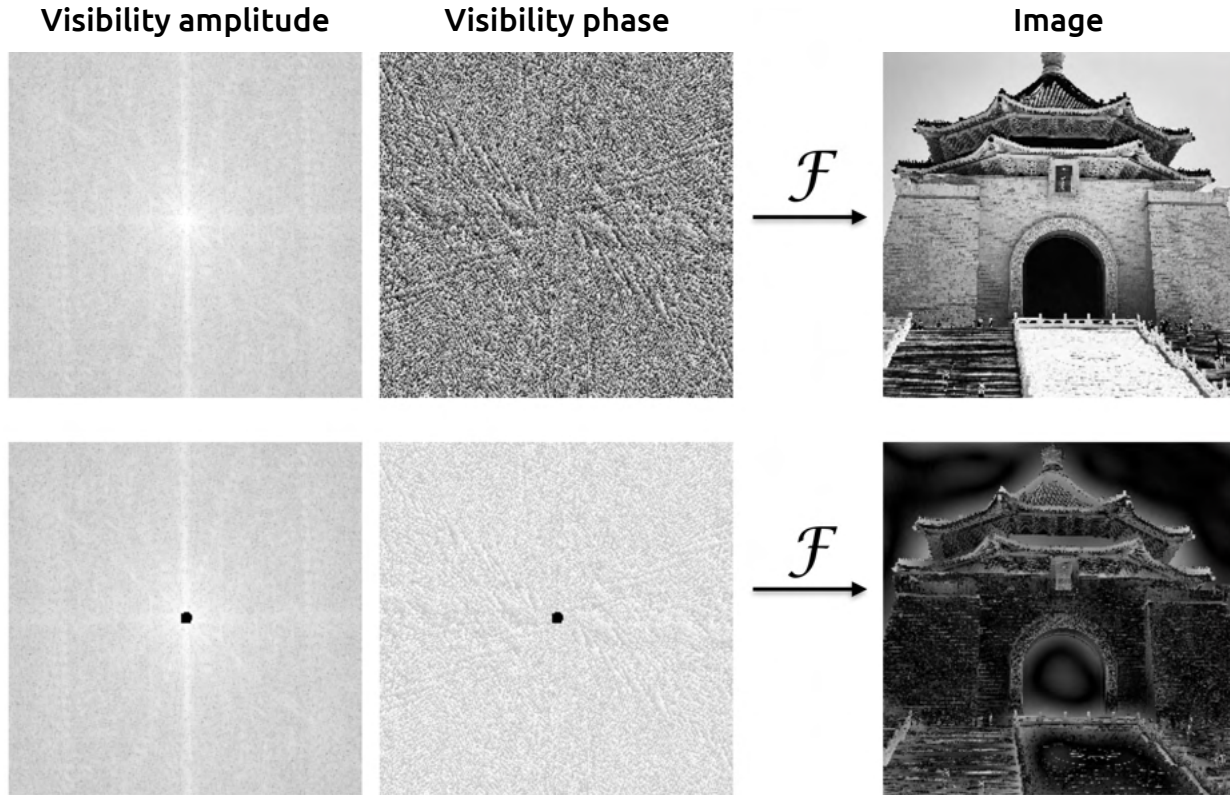
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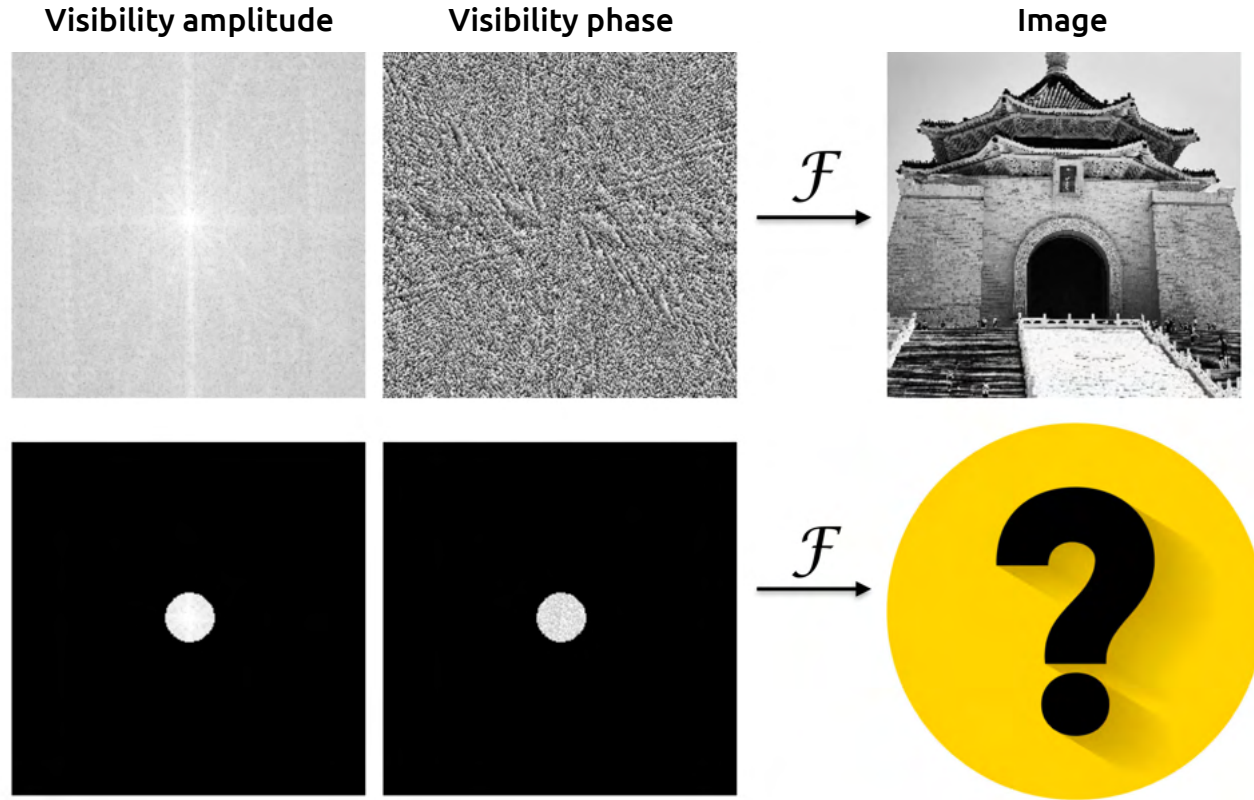
Impact of Finite uv -Sampling



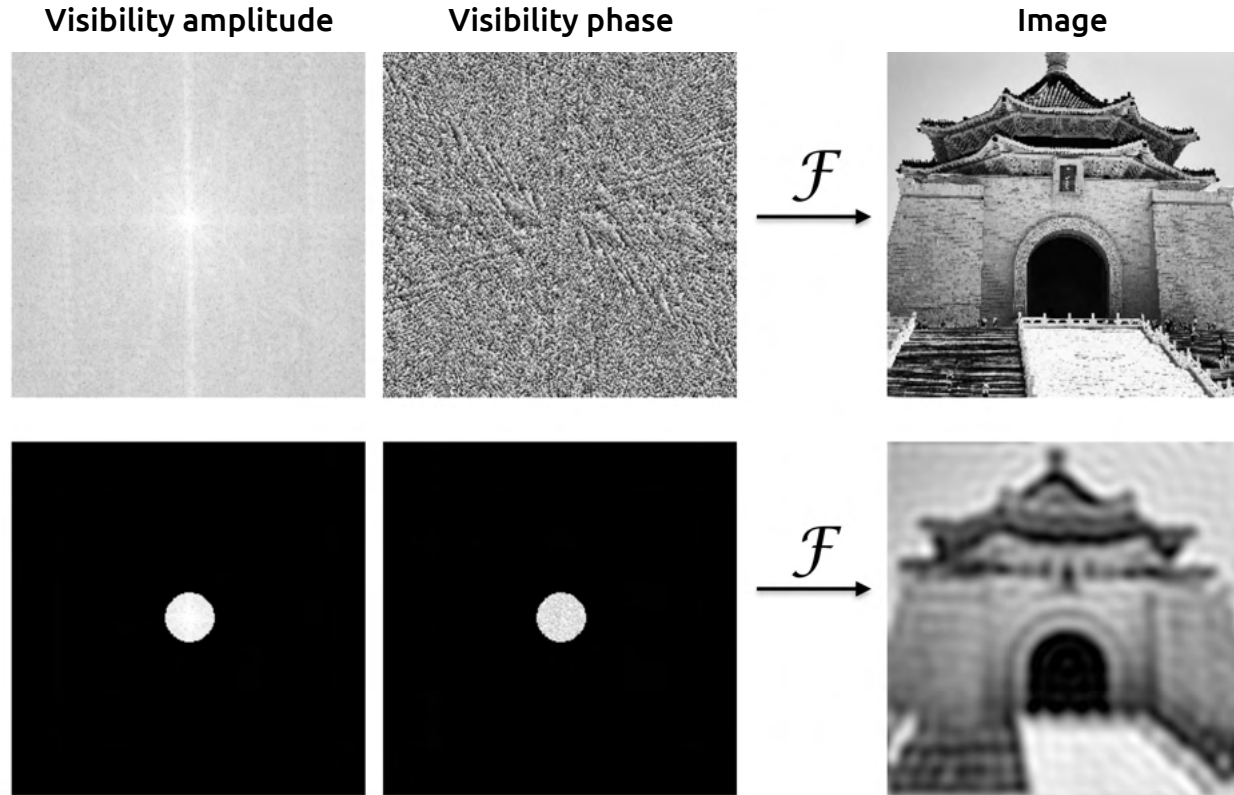
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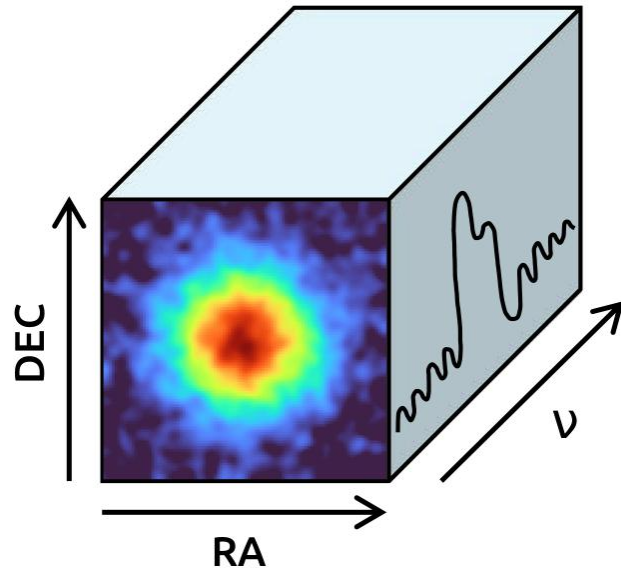


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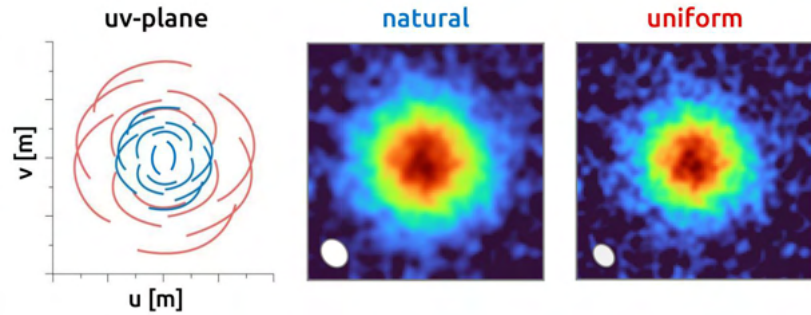


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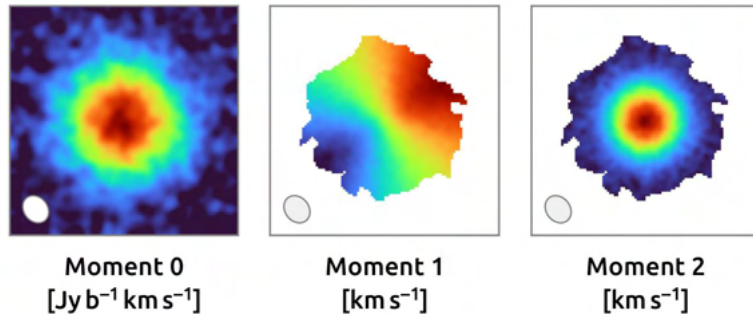


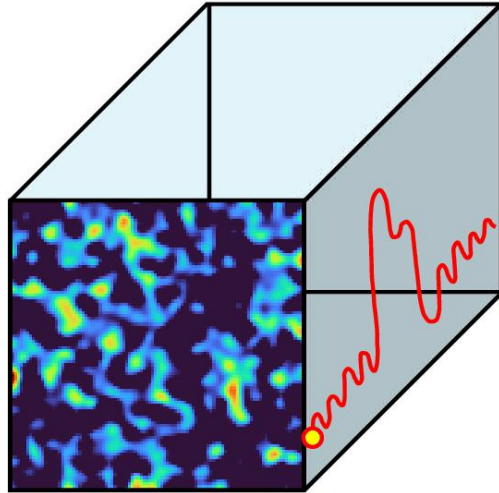


► Different data weighting schemes



► Spectral line moment maps

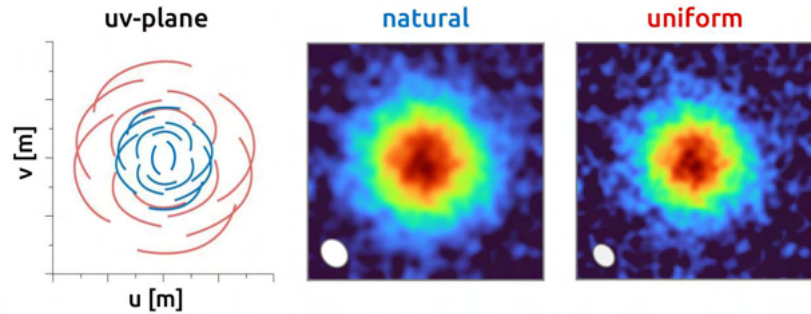




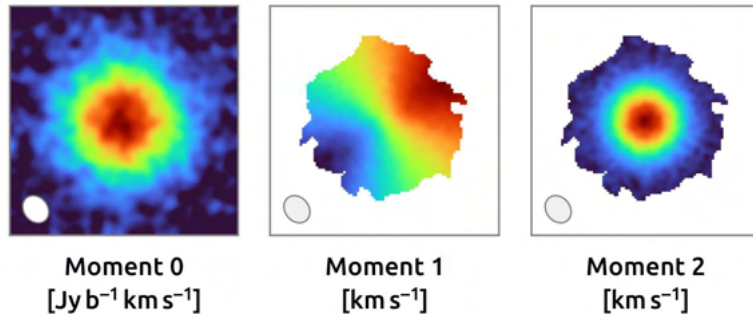
ALMA-related (animated) figures:

<https://drive.google.com/drive/folders/1JlXHTZqovMDQcEMiH5vDaevE4u5Wvzz0>

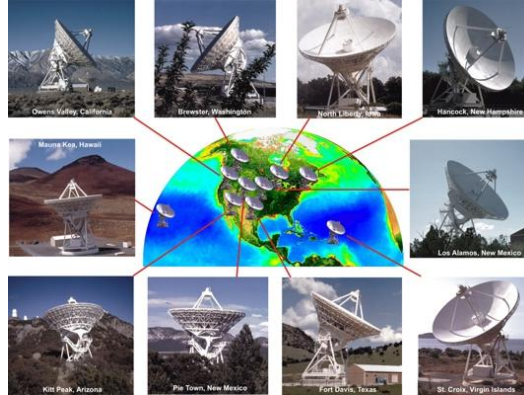
► Different data weighting schemes



► Spectral line moment maps



Northern Hemisphere Interferometers



VLBA (10 antennas), mainly USA



LOFAR (~50 stations)
Netherlands

LOFAR → meters (MHz)
VLA → cm (GHz)
VLBA → cm (GHz)
NOEMA → (sub)mm
(70–375 GHz)

Southern Hemisphere Interferometers



ALMA → (sub)mm
(35–950 GHz)

ATCA → cm/mm
(1.1–105 GHz)

MeerKAT → cm
(0.5–3.5 GHz)



MeerKAT (64 antennas)
South Africa

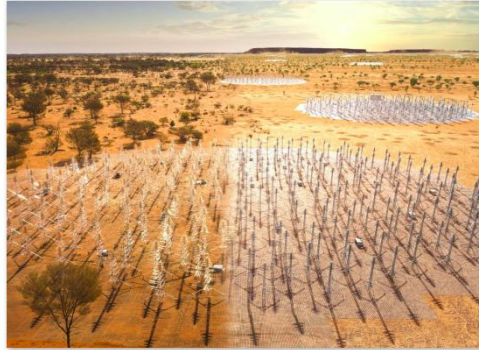


ATCA (6 antennas)
Australia

Next-Generation Interferometers



Square Kilometre Array Observatory

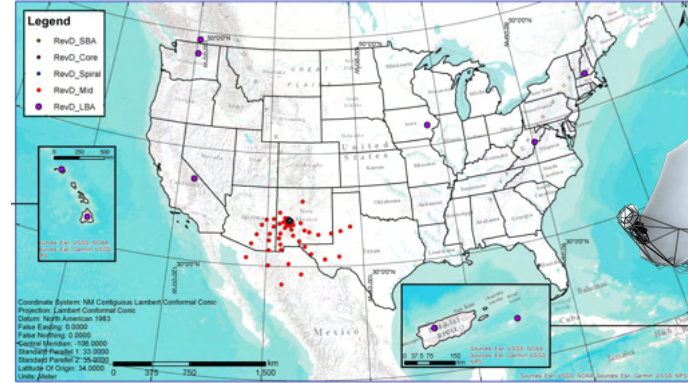


Extreme sensitivity to faint, large-scale emission

- **SKA-Low** (Australia): 50 – 350 MHz (meters)
- **SKA-Mid** (South Africa): 0.35 – 15 GHz (cm)
- Largest collecting area ever built: **1 million m² of collecting area** to study critical scientific questions, from the birth of our Universe to the origins of life.



Next Generation VLA (ngVLA)



High-reso imaging bridging VLA and ALMA

- **1.2 – 116 GHz** Frequency Coverage
- **Array Design:** 244 x 18m offset Gregorian Antennas
 - **Core:** 114 fixed antennas; $B_{\max} = 4.3$ km
 - **Spiral:** 54 fixed antennas; $B_{\max} = 39$ km
 - **Mid:** 46 fixed antennas spread into NM, AZ, TX, MX; $B_{\max} = 1070$ km
 - **Long:** 30 x 18m antennas located across continent; $B_{\max} = 8860$ km
- **Short Baseline Array:** 19 x 6m offset Gregorian Antennas
 - Use 4 x 18m in **TP mode** to fill in (u, v) hole.